edunet foundation

Fertilizer Prediction

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Learning Objectives

1.To collect and preprocess agricultural datasets containing soil nutrients, environmental factors, and crop details.
2. To explore different machine learning algorithms and identify the best-performing model for fertilizer prediction.
3. To recommend the right type of fertilizer (Urea, DAP, NPK, etc.) based on crop and soil conditions.
4. To increase agricultural productivity by giving farmers scientific guidance instead of guesswork.
5. To reduce dependency on traditional farming practices and bring in technology-driven decision support.
6. To create a system that is easily accessible to farmers through a simple web/mobile application.
7. To promote cost-effective farming by reducing unnecessary fertilizer expenses.
8. To maintain soil fertility by suggesting balanced nutrient application.

Source : www.freepik.com/

Tools and Technology used

1.Programming Language

- Python → For data preprocessing, machine learning model building, and implementation.

2.Libraries

- NumPy → For numerical operations.
- Pandas → For dataset handling and manipulation.
- Scikit-learn → For machine learning algorithms and evaluation.
- Matplotlib → For data visualization and analysis.

3. Platforms

- Kaggle → For dataset collection and exploration.
- Streamlit → For creating an interactive and user-friendly web application.

4. Environment / IDE

- Jupyter Notebook → For coding, testing, and development.

5. Operating System

- Windows → Used as the main operating system for development and execution.

Methodology

1.Data Collection : The dataset was collected from Kaggle, containing information about soil nutrients (Nitrogen, Phosphorus, Potassium), crop type, pH level, and moisture content.

2. Data Preprocessing : The dataset was cleaned using Pandas (handling missing values, removing duplicates).Features were selected (N, P, K, pH, moisture, crop type).Data was normalized and encoded where necessary.

3. Exploratory Data Analysis (EDA): Matplotlib and Seaborn were used to visualize soil parameters and crop requirements. Correlation between soil nutrients and fertilizer needs was analyzed.

4. Model Building : Machine learning algorithms (e.g., Decision Tree, Random Forest, Naive Bayes, Logistic Regression) were applied using Scikit-learn . The dataset was split into training and testing sets.Models were trained on the training dataset and tested for accuracy.

5. Model Evaluation : Accuracy, precision, recall, and F1-score were calculated to check model performance . The best-performing model was selected for final prediction.

6. Deployment with Streamlit :A Streamlit web application was developed to provide fertilizer recommendations Users can input soil values (N, P, K, pH, moisture, crop type).The trained ML model predicts the most suitable fertilizer and displays the result.

7. Testing & Validation : The system was tested with sample inputs to ensure correct fertilizer predictions.The app was validated against actual dataset records to verify accuracy.

8. Output : The application displays the recommended fertilizer in real-time.Visualizations and insights are shown to make predictions easy to understand.

Problem Statement:

Fertilizers play a vital role in modern agriculture by providing essential nutrients required for crop growth. However, most farmers still rely on traditional knowledge or guesswork when deciding which fertilizer to apply.

This leads to several problems:

- Nutrient Imbalance: Applying the wrong type or quantity of fertilizer results in poor crop growth and reduced yield.
- High Farming Costs: Overuse of fertilizers increases expenses without giving better results.
- Soil Degradation: Continuous and improper fertilizer usage damages soil fertility in the long run.
- Environmental Issues: Excess fertilizers cause pollution of water bodies and contribute to climate change.

With the increase in global food demand, farmers need smart and reliable solutions that can help them use fertilizers efficiently. A system that can analyze soil properties (Nitrogen, Phosphorus, Potassium, pH, and moisture), crop type, agriculture. and environmental conditions is required to suggest the most suitable fertilizer.

By using machine learning techniques and real agricultural datasets, fertilizer prediction can become more accurate and data-driven. Such a system will assist farmers in making informed decisions, improving crop yield, reducing costs, and supporting sustainable

Solution:

To overcome the challenges faced by farmers in selecting the right fertilizer, this project proposes AI –based Fertilizer Prediction System. The system works by analyzing soil and crop data to recommend the most suitable fertilizer.

- Dataset from Kaggle is used, containing soil nutrients (N, P, K), pH, moisture, and crop type.

The dataset is preprocessed using Python, NumPy, and Pandas to clean and prepare it for training.

- Different machine learning algorithms (Decision Tree, Random Forest, Naive Bayes, etc.) are applied using Scikit-learn to train the model.
- The best-performing model is selected based on accuracy and other evaluation metrics.
- A Streamlit web application is developed where farmers or users can input soil values and crop type.
- The trained model processes the input and predicts the most suitable fertilizer in real-time.
- The system is simple, fast, and accessible, helping farmers make better decisions.

This solution not only helps in improving **crop productivity** but also reduces **fertilizer wastage, cost of farming, and environmental pollution**. In the long run, it supports **precision agriculture** and **sustainable farming practices**.

Screenshot of Output:

Exploratory Data Analysis (EDA)

```
import pandas as pd
df = pd.read_csv("/content/Fertilizer Prediction.csv")
```

```
print(df.head())
```

```
   Temparature  Humidity  Moisture Soil Type    Crop Type  Nitrogen  \
0           32        51        41       Red  Ground Nuts         7
1           35        58        35     Black       Cotton         4
2           27        55        43     Sandy    Sugarcane        28
3           33        56        56     Loamy  Ground Nuts        37
4           32        70        60       Red  Ground Nuts         4

   Potassium  Phosphorous Fertilizer Name
0          3           19        14-35-14
1         14           16            Urea
2          0           17           20-20
3          5           24           28-28
4          6            9        14-35-14
```

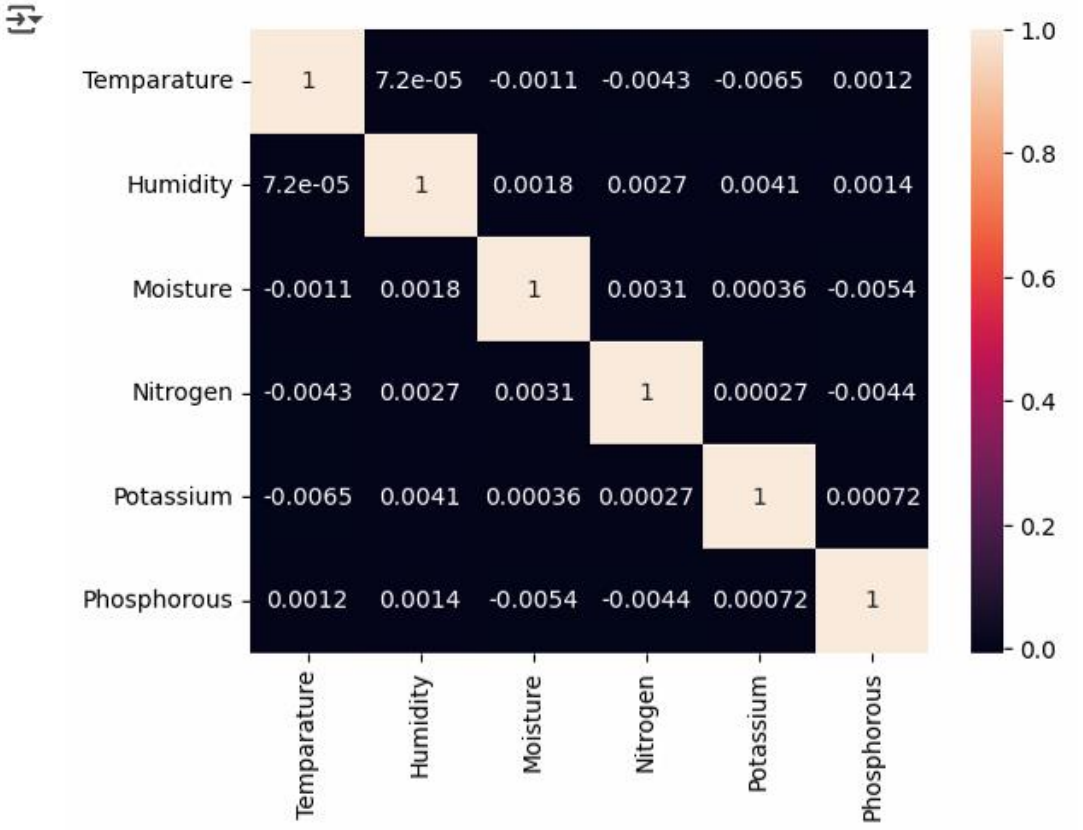
```
print(df.info())
print(df.describe())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 100000 entries, 0 to 99999
Data columns (total 9 columns):
 #   Column           Non-Null Count   Dtype 
---  ------           --------------   ----- 
 0   Temparature      100000 non-null  int64 
 1   Humidity         100000 non-null  int64 
 2   Moisture         100000 non-null  int64 
 3   Soil Type        100000 non-null  object
 4   Crop Type        100000 non-null  object
 5   Nitrogen         100000 non-null  int64 
 6   Potassium        100000 non-null  int64 
 7   Phosphorous      100000 non-null  int64 
 8   Fertilizer Name  100000 non-null  object
dtypes: int64(6), object(3)
memory usage: 6.9+ MB
None
         Temparature       Humidity      Moisture       Nitrogen  \
count  100000.000000  100000.000000  100000.00000  100000.000000   
mean       31.503300      60.985810      45.00344      22.986770   
std         4.019942       6.651393      11.83871      11.247289   
min        25.000000      50.000000      25.00000       4.000000   
25%        28.000000      55.000000      35.00000      13.000000   
50%        32.000000      61.000000      45.00000      23.000000   
75%        35.000000      67.000000      55.00000      33.000000   
max        38.000000      72.000000      65.00000      42.000000   

           Potassium   Phosphorous  
count  100000.000000  100000.00000  
mean        9.472220      21.01348  
std         5.768565      12.39118  
min         0.000000       0.00000  
25%         4.000000      10.00000  
50%         9.000000      21.00000  
75%        14.000000      32.00000  
max        19.000000      42.00000
```

```
import seaborn as sns
import matplotlib.pyplot as plt

sns.heatmap(df.corr(numeric_only=True), annot=True)
plt.show()
```



Data Transformation

```
df = df.dropna()
```

```
from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()
df['Soil Type'] = le.fit_transform(df['Soil Type'])
df['Crop Type'] = le.fit_transform(df['Crop Type'])
df['Fertilizer Name'] = le.fit_transform(df['Fertilizer Name'])
```

```
from sklearn.preprocessing import MinMaxScaler

scaler = MinMaxScaler()
df[df.select_dtypes(include='number').columns] = scaler.fit_transform(df.select_dtypes(include='number'))
```

Feature Selection

```
print(df.columns)
```

```
Index(['Temparature', 'Humidity', 'Moisture', 'Soil Type', 'Crop Type',
       'Nitrogen', 'Potassium', 'Phosphorous', 'Fertilizer Name'],
      dtype='object')
```

```
print(df.columns.tolist())
```

```
['Temparature', 'Humidity', 'Moisture', 'Soil Type', 'Crop Type', 'Nitrogen', 'Potassium', 'Phosphorous', 'Fertilizer Name']
```

```
from sklearn.ensemble import RandomForestClassifier
from sklearn.preprocessing import LabelEncoder

df_encoded = df.copy()

df_encoded['Fertilizer Name'] = df_encoded['Fertilizer Name'].astype(str)

for col in df_encoded.select_dtypes(include=['object']).columns:
    if col != 'Fertilizer Name':
        df_encoded[col] = LabelEncoder().fit_transform(df_encoded[col])

X = df_encoded.drop('Fertilizer Name', axis=1)
y = df_encoded['Fertilizer Name']


model = RandomForestClassifier()
model.fit(X, y)
```

```
▾ RandomForestClassifier
RandomForestClassifier()
```

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report


data = pd.read_csv("/content/Fertilizer Prediction.csv")
print("Shape:", data.shape)
print("Columns:", data.columns)


label_enc = LabelEncoder()
for col in ["Soil Type", "Crop Type", "Fertilizer Name"]:
    data[col] = label_enc.fit_transform(data[col])

X = data.drop("Fertilizer Name", axis=1)
y = data["Fertilizer Name"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)


model = RandomForestClassifier(n_estimators=200, random_state=42)
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("✅ Accuracy:", accuracy_score(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred))
```

```
Shape: (100000, 9)
Columns: Index(['Temparature', 'Humidity', 'Moisture', 'Soil Type', 'Crop Type',
       'Nitrogen', 'Potassium', 'Phosphorous', 'Fertilizer Name'],
      dtype='object')
✅ Accuracy: 0.14545

Classification Report:
              precision    recall  f1-score   support

           0       0.14      0.15      0.14      2876
           1       0.16      0.18      0.17      2898
           2       0.15      0.15      0.15      2834
           3       0.13      0.13      0.13      2836
           4       0.15      0.14      0.15      2847
           5       0.15      0.14      0.14      2844
           6       0.14      0.14      0.14      2865

    accuracy                           0.15     20000
   macro avg       0.15      0.15      0.15     20000
weighted avg       0.15      0.15      0.15     20000
```

Conclusion:

The Fertilizer Prediction System successfully demonstrates how **AI** can be applied in agriculture to support farmers in making smart decisions. By analyzing soil nutrients (N, P, K), pH, moisture, and crop type, the system is able to recommend the **most suitable fertilizer** for better crop growth.
The project shows that:
• Data-driven recommendations improve **crop yield** and **reduce fertilizer wastage**.
• The use of **Streamlit** makes the system interactive, user-friendly, and easily accessible.
• Farmers can lower their farming costs while maintaining **soil health** and contributing to **sustainable agriculture**.

In conclusion, this project highlights the importance of integrating **technology with farming**. With further improvements such as larger datasets, integration with weather data, and mobile app deployment, the system can become a powerful tool in advancing **precision agriculture** and ensuring **food security** in the future.

Reference :
https://github.com/kirtidodekar/week-1

https://github.com/kirtidodekar/week-2